Experiment:- Study of LAN Topology, Network devices and Address configuration.

Objective:-

1) To study the LAN topology used in collage campus.
2) To identify networking device & understand their function.
3) To analyze wired & wireless communication method used in the network.
4) To measure speed of Network for wired & wireless connections.
5) To obtained MAC, IP Address & subnet, mask.

Problem Statment - 1

- Draw and identify LAN topology of collage campus & clearly label all components such as routes, switches & etc.

Soln:- The collage campus network observed this study consists of:

* Layer - 3 core switches in the server room.
* Managed by POE + switches
* Wireless Access point (APs)
* Eathernet port in class room & laboratories.

Network Topology:- Hybrid hierarchical topology which combines:-

i) Tree Topology
ii) Extended Star Topology
iii) ~~Rtg~~ Ring Topology (server room)

The network Hierarchical layer :-

① Core layer
② Distribution layer
③ Access layer

* Core Layer :-

↳ Core layer contains server room and server room have ~~firew~~ Firewall, Router, 2-L3+ Switches, patch Rack, server, inverter etc

* Distribution layer :-

↳ Distribution layer contains 4- switches.

↳ The four switches are

① → Main Building → R7 (Room no. 213, 2nd floor)
2 → Main Building → R1 (Room no. 301, 3rd floor)
3 → Main Building → R6 (Room no. 313, 1st floor)
4 → Main Building → R2 (Room no 201, 2nd floor)

↳ This 4-switches contains :-

① 18 Port Gigabit Easy smart switch with 16- port. TL-SG1218MPE

② SG2418XMP → 24 Port Giga bit & 4 port 10GESFP+, L2+ Managed switches with 24 Port POE+.

↳ TP- Link TL-SG1218MPE :-

* Likely used for: CCTV camera, wifi access point etc.
* 18 port gigabit "Easy Smart" switch.
* 16 Port support POE+

② TP-Link Omada SG3428XMP

* Higher-end managed layer 2+ switch.
* 24 POE+ Port, uplinks, 10GSFP
* Usually act as a main distribution for each floor.

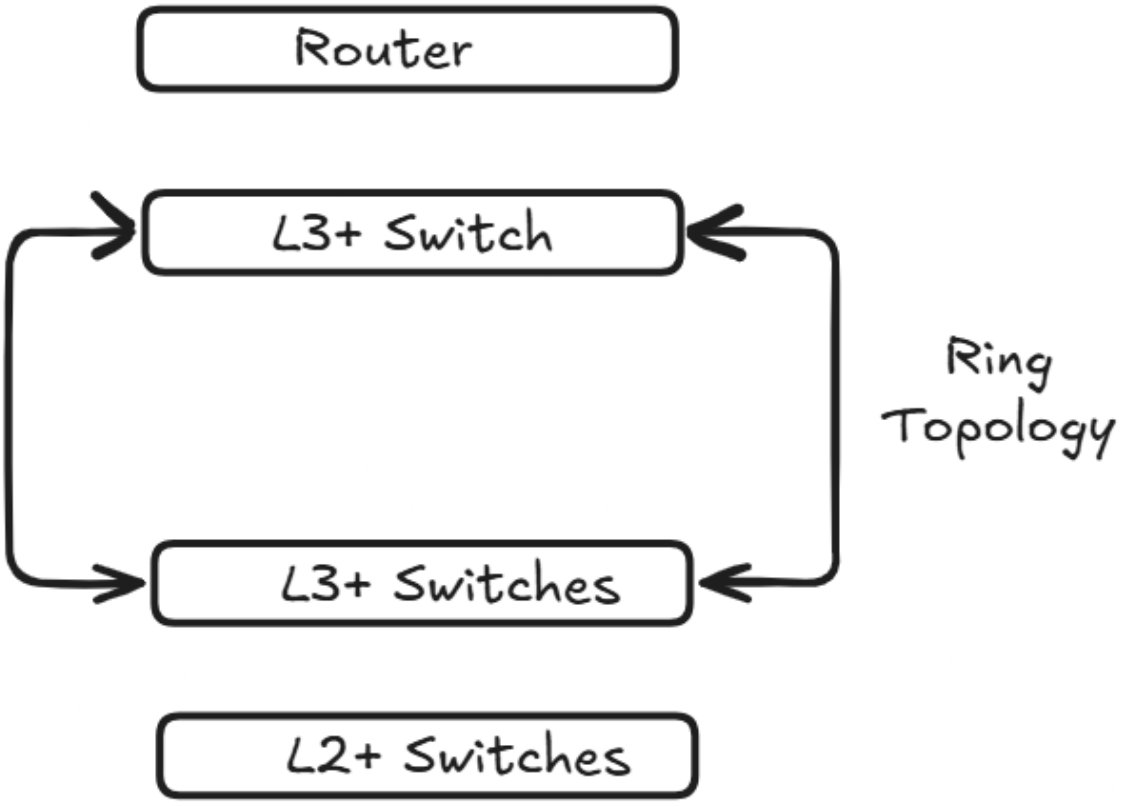
* Access Layer

↳ Access layer contains 3 switches i.e.
1) Main Building R1 - [Room no - 301, 3rd floor)
2) Main Building R4 - [Roon no 05, Ground floor)
3) Main Building R5 - [Room no. 18, ground floor)

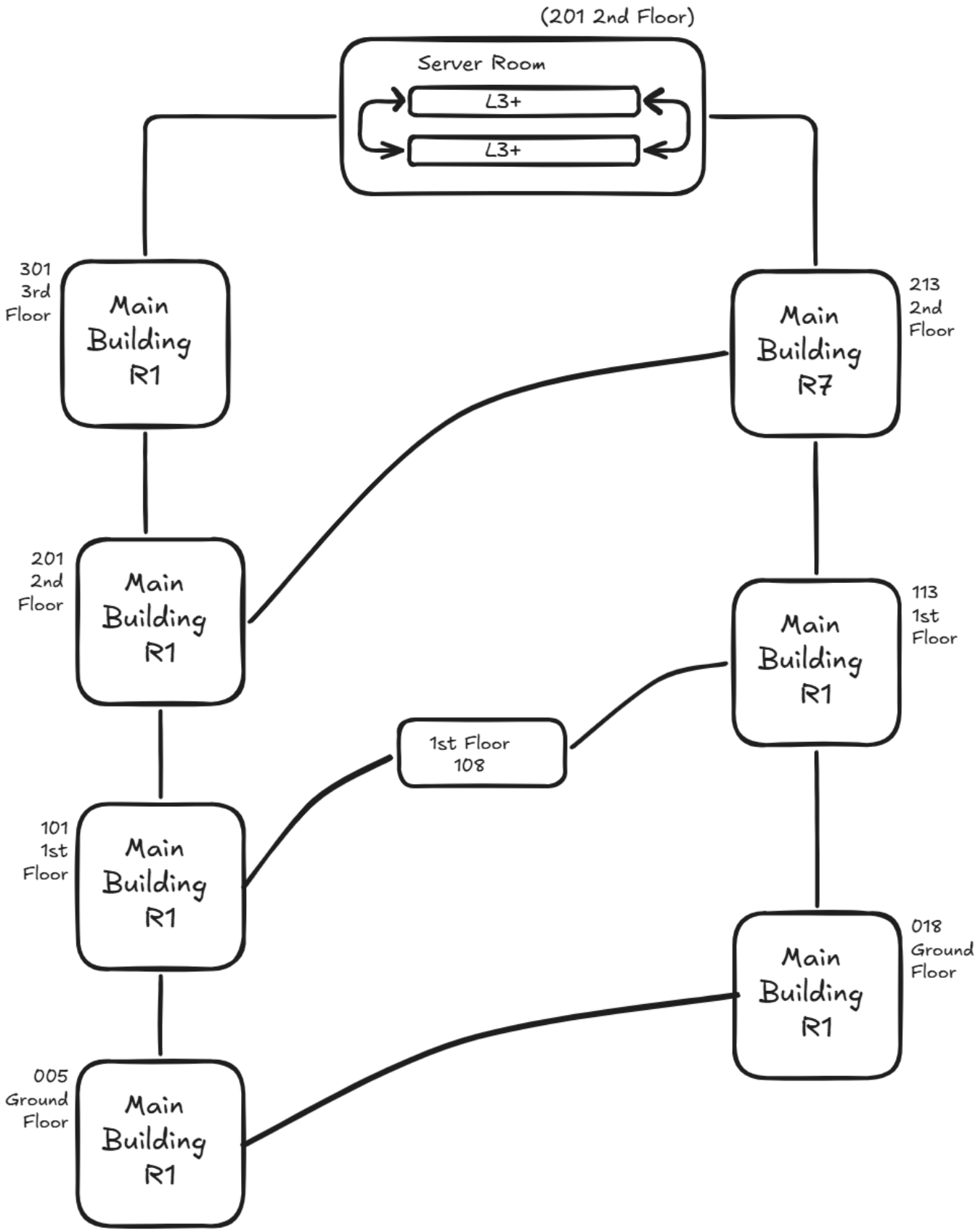
↳ Every switches act as access layer which helps to provide network connectivity to all classroom, lab, faculty room, etc.

↳ Access layer directly connect :-
① Ethernet Port.
② Wireless AP
③ Lab system
④ Outdoor wireless AP (ex-018)

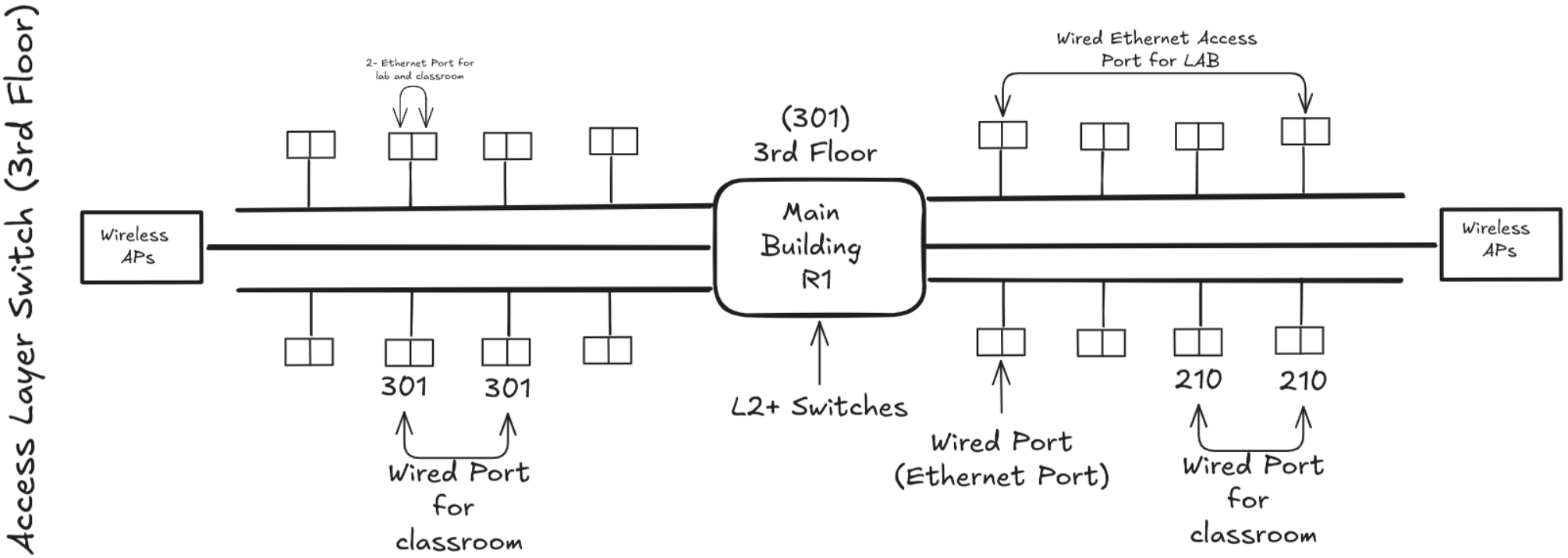
Server Room Architecture/ Topology



Topology



-> R1, R7, R2, R6 -> Distribution Layer
-> Server room -> Core Layer
-> R1, R2, R3, R4, R5, R6, R7 -> Access Layer


Access Layer Switch (3rd Floor)
2- Ethernet Port for lab and classroom
Wireless APs
(301) 3rd Floor
Main Building R1
Wired Ethernet Access Port for LAB
Wireless APs
301
301
Wired Port for classroom
L2+ Switches
Wired Port (Ethernet Port)
210
210
Wired Port for classroom

Wireless Connection

```
PS C:\Users\aksha> ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

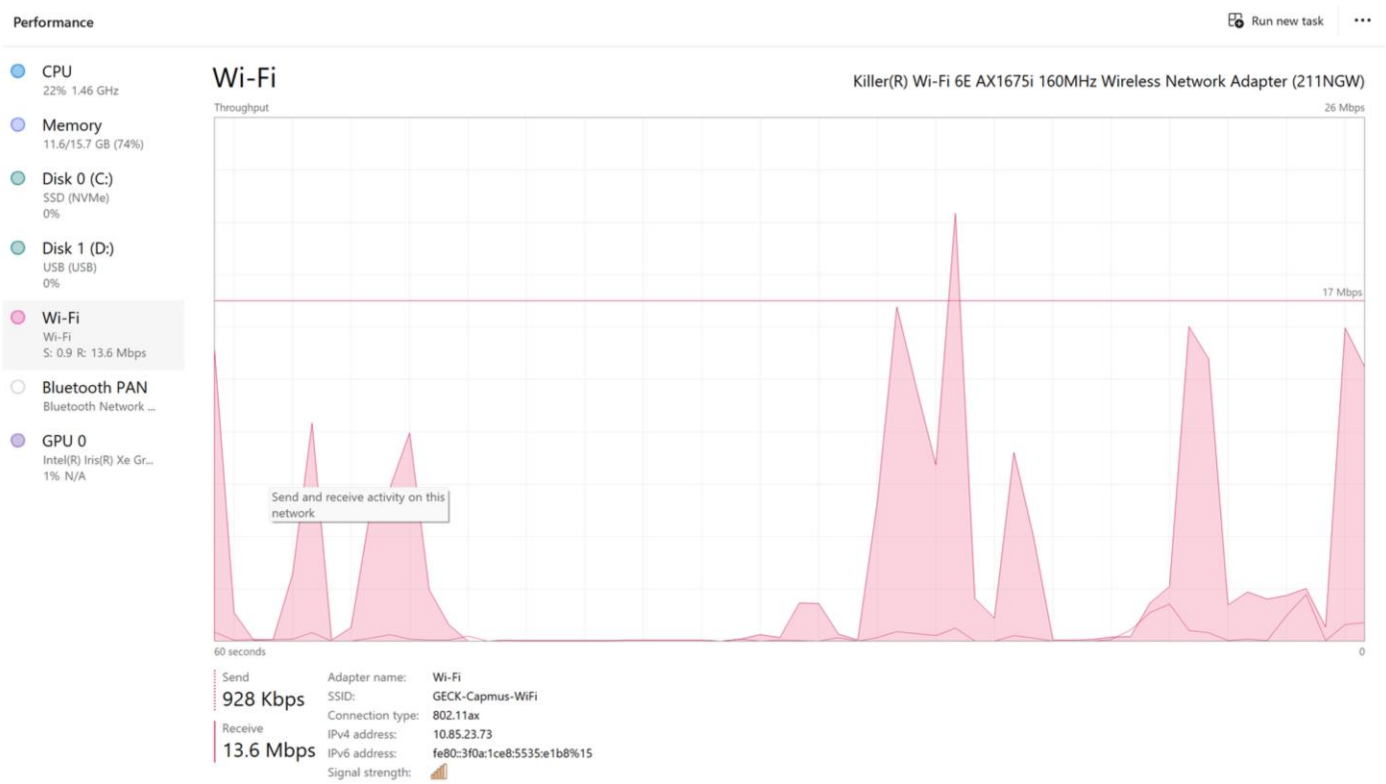
   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::3f0a:1ce8:5535:e1b8%15
   IPv4 Address. . . . . . . . . . . : 10.85.23.73
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Default Gateway . . . . . . . . . : 10.85.16.1

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
PS C:\Users\aksha>
```

Using "ipconfig"

Wireless Speed




```
PS C:\Users\aksha> getmac

Physical Address    Transport Name
=================== ============================================================
FA-E1-97-14-C9-AF   \Device\Tcpip_{A10F60CA-B057-4077-9415-3D8583412623}
A0-02-A5-CB-A3-03   Media disconnected
PS C:\Users\aksha>
```

-> MAC Address
-> using "getmac"

Wired Connection

```
PS C:\Users\aksha> ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2409:40e4:16:15ec:6226:a2d2:4037:a24d
   Temporary IPv6 Address. . . . . . : 2409:40e4:16:15ec:124:6cd8:50a6:9016
   Link-local IPv6 Address . . . . . : fe80::a403:5015:36c7:cfd2%12
   IPv4 Address. . . . . . . . . . . : 10.60.18.56
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::c54:b8ff:feb8:4eb2%12
                                       10.60.18.98

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
PS C:\Users\aksha>
```

Using "ipconfig"

Difference b/w wired & wireless speed?

Soln:

Wired Network	Wireless Network
↳ Faster Communication	↳ Slower Communication
↳ Stable Connection	↳ Affected by distance/ interference
↳ low latency	↳ Higher latency
↳ Dedicated physical medium	↳ Shared radio medium

→ Ethernet provides direct physical communic-ation using network cables, resulting in higher speed & stable performance.

→ Wireless connection use radio waves which are affected by interference, distance, obstacles & multiple connected users.

8) Role of subnet mask in network?

Soln) A subnet mask identifies the network portion & host portion of an IP address. It helps ~~def~~ devices determine whether another ~~ehat~~ device belong to the same network or different.

* It is important for :-
  1. Network Segment
  2. Efficient Routing
  3. Communication b/w devices.

→ Observation

During the study of the collage networ infrastruct-ure :-

1. A centralised server room was used as network backbone.
2. Two L3 switches were connected in redundant cyclic structure.
3. Managed TP-Link POE+ switches were used in network cabinets

→ Advantage of Existing Network ?

* Scalable achitecture
* Centralized management
* Supports POE+ devices
* Easy expansion of floors & classrooms.
* Supports both wireless & wired.
* Redundand link for failower.